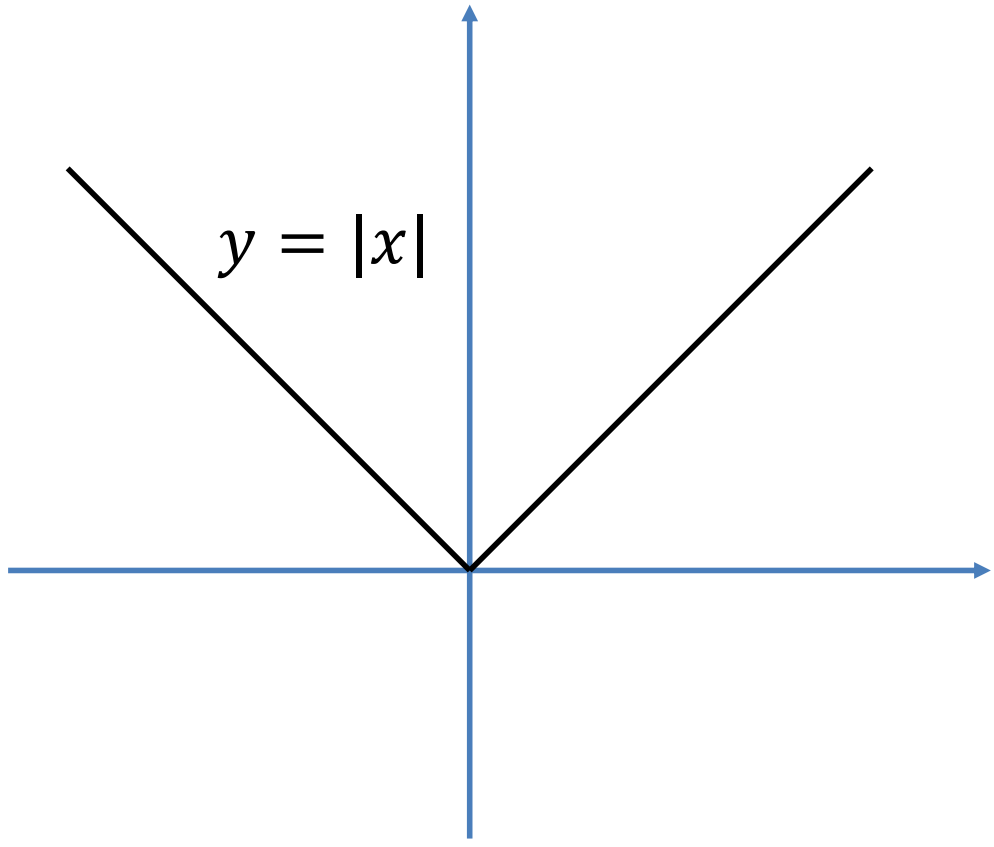
2.4 Piecewise-defined Functions

Recall the absolute value function $|x| = \begin{cases} -x & \text{if } x < 0, \\ x & \text{if } x \geq 0. \end{cases}$

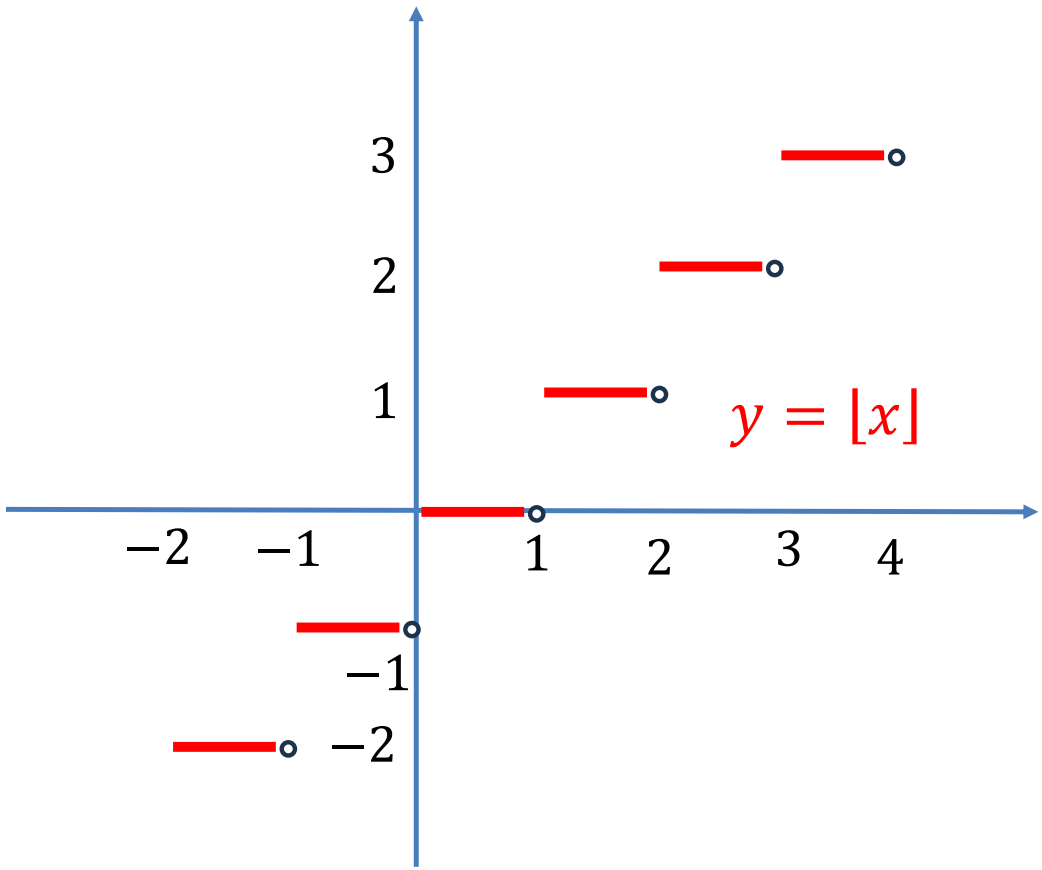
It is an example of *piecewise-defined functions*. Its graph is shown below (two linear pieces).



2.4 Piecewise-defined Functions

Recall that $\lfloor x \rfloor$ denotes the nearest integer not exceeding x (round x down). This is called the *floor* function, which can be expressed as a piecewise-defined function (with many pieces):

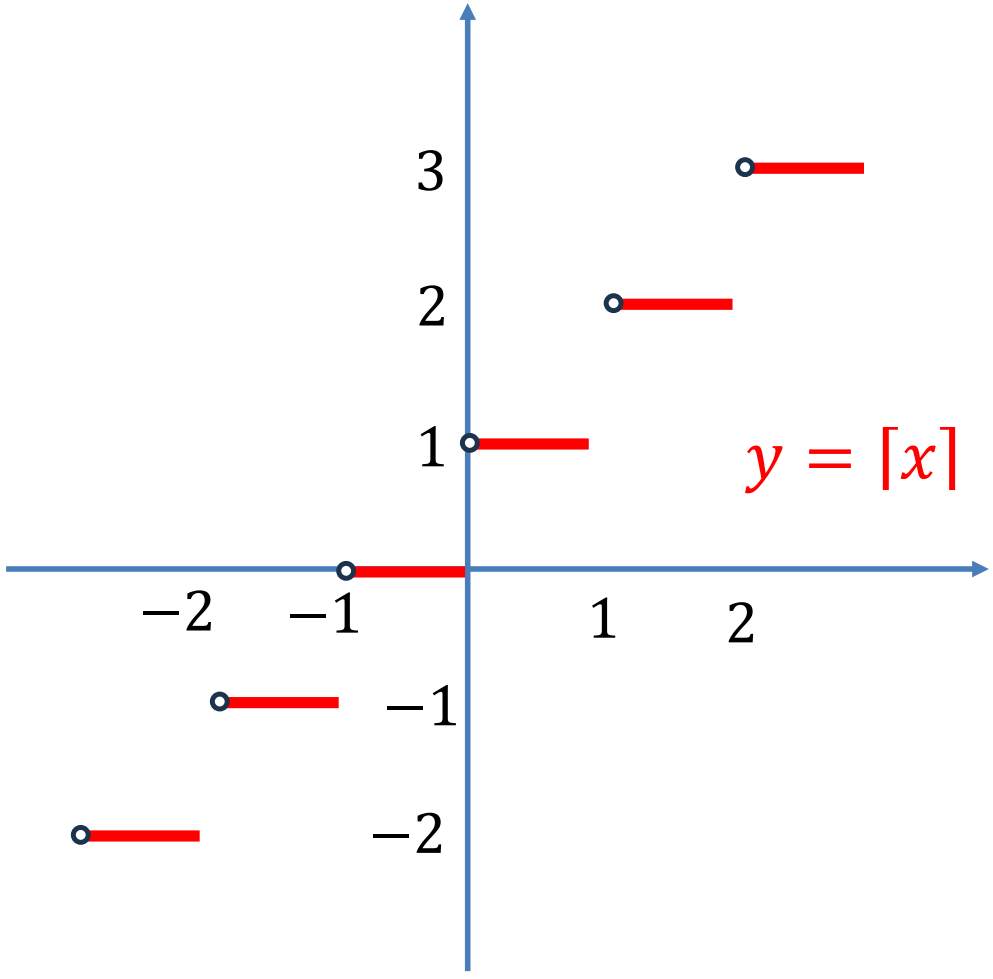
$$\lfloor x \rfloor = n : n \leq x < n + 1,\ n \text{ is any integer.}$$



2.4 Piecewise-defined Functions

The ceiling function $\lceil x \rceil$ (round x up) can also be expressed as a piecewise-defined function:

$$\lceil x \rceil = n : n < x \leq n + 1,\ n \text{ is any integer.}$$



2.4 Power Functions

A function of the form $f(x) = x^r$, where r is a real number, is called a power function.

Example The following are all power functions:

$$f(x) = x^3, \;\; g(x) = x^{1/3}, \;\; h(x) = x^{1/2}, \;\; q(x) = x^{-3}.$$

We note

$$x^{1/2} = \sqrt{x}, \quad x^{1/3} = \sqrt[3]{x},, \quad x^{-3} = \frac{1}{x^3},$$

$$Dom\{f\} = (-\infty, \infty),$$
$$Dom\{g\} = (-\infty, \infty),$$
$$Dom\{h\} = [0, \infty),$$
$$Dom\{q\} = (-\infty, 0) \cup (0, \infty).$$

2.4 Power Functions

Example The function $h(x) = \sqrt{x+2} = (x+2)^{1/2}$ has domain

$$Dom\{h\} = \{x \in \mathbb{R} : x + 2 \geq 0\} = [-2, \infty).$$

It can be expressed as $f \circ g$, where

$$f(x) = x^{1/2}, \quad g(x) = x + 2.$$

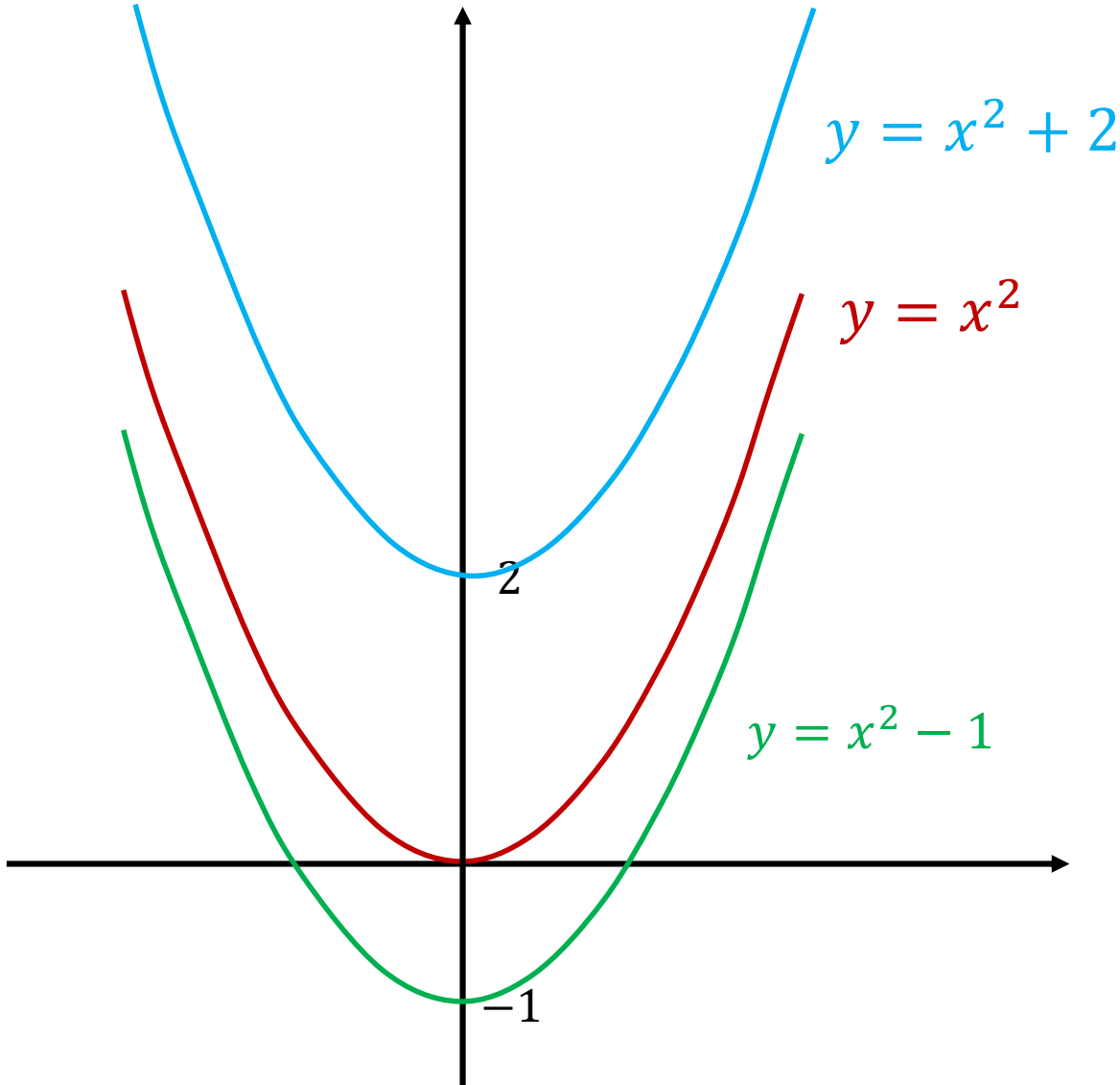
Example The function $h(t) = \dfrac{1}{\sqrt{t+2}} = (t+2)^{-1/2}$ has domain

$$Dom\{h\} = \{t \in \mathbb{R} : t + 2 > 0\} = (-2, \infty).$$

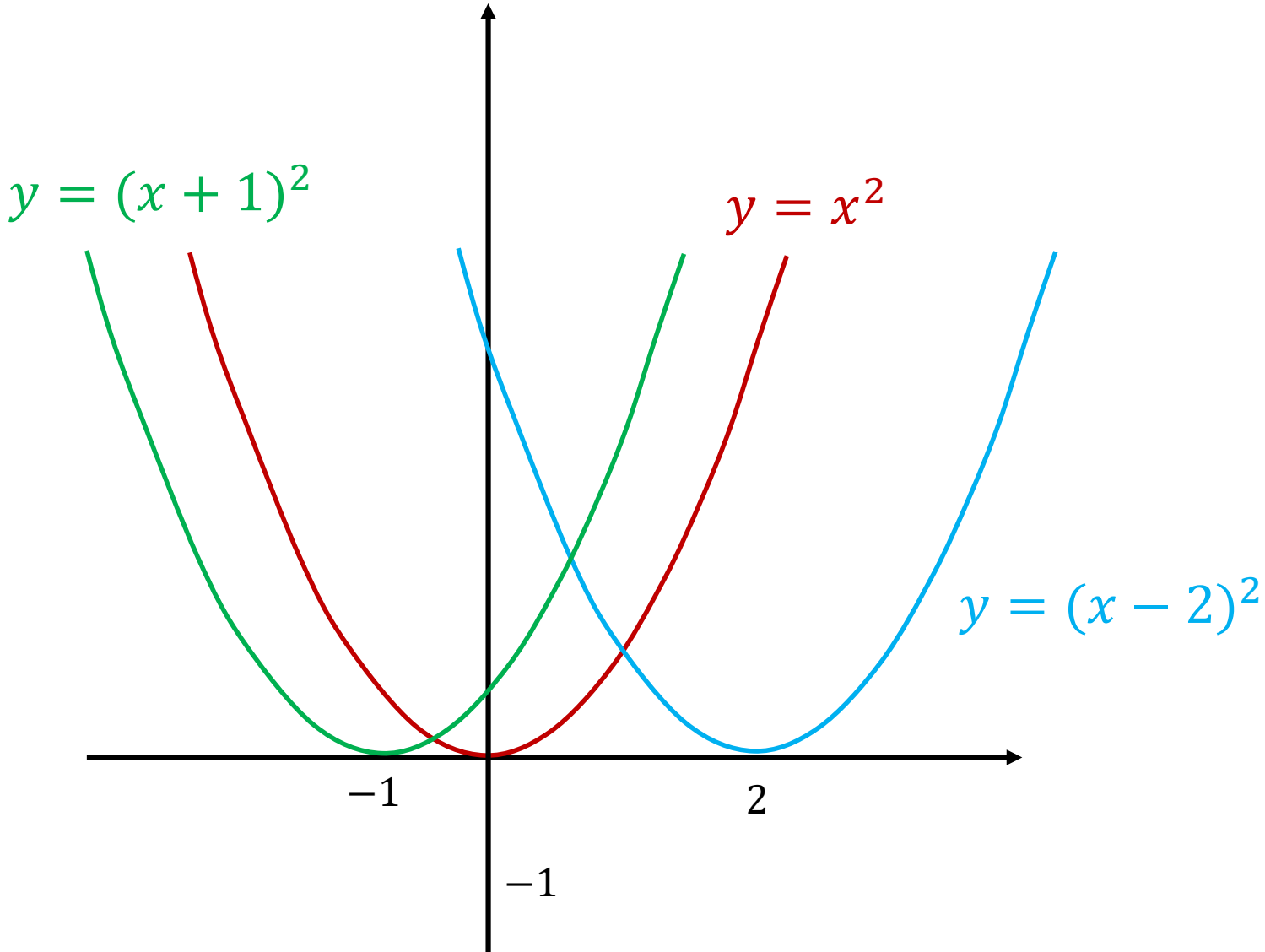
It can be expressed as $f \circ g$, where

$$f(t) = t^{-1/2}, \quad g(t) = t + 2.$$

2.4 Vertical Shift of Graphs



2.4 Horizontal Shift of Graphs



2.4 Vertical and Horizontal Shifts

Let G be the graph of a function $f(x)$. Let c be a positive constant. To obtain the graph of

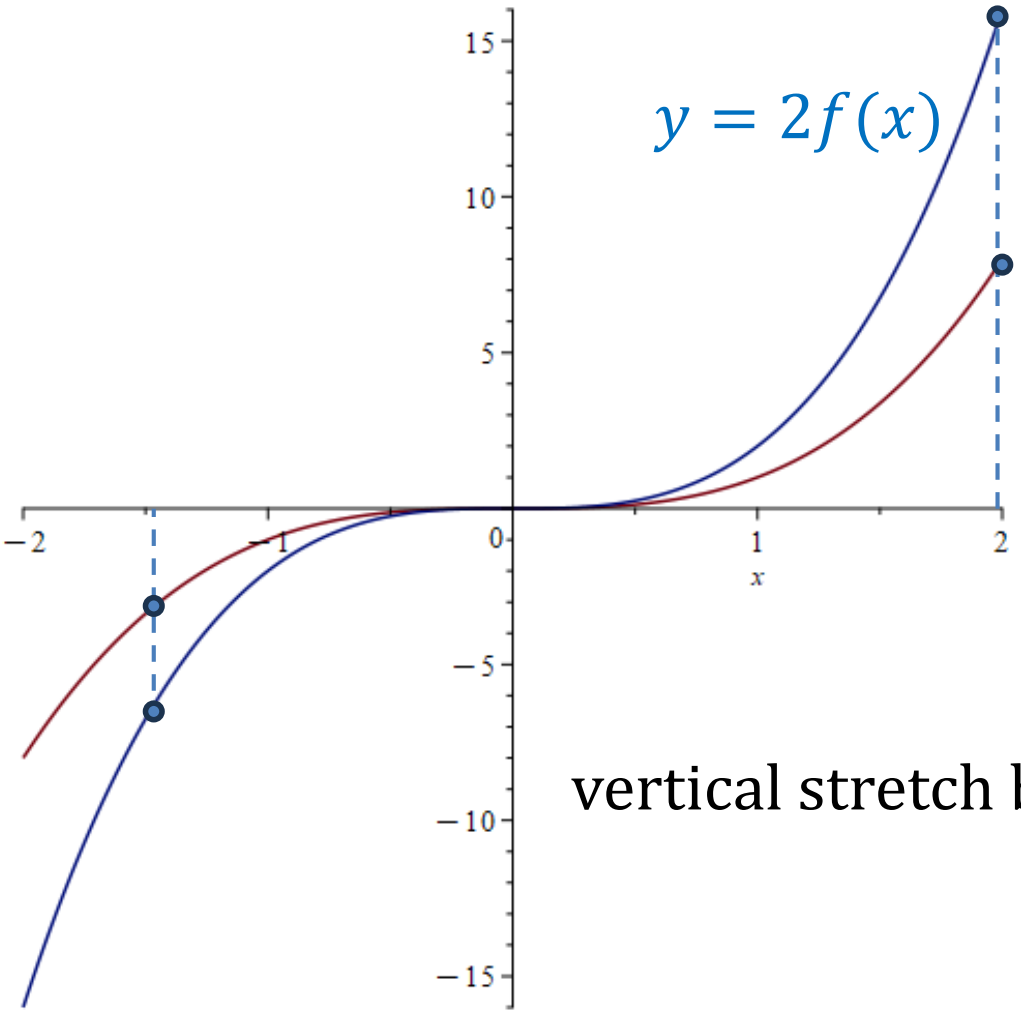
- $f(x) + c$, shift G up by c units;
- $f(x) - c$, shift G down by c units ;
- $f(x - c)$, shift G to the right by c units;
- $f(x + c)$, shift G to the left by c units.

2.4 Vertical and Horizontal Stretch/Compress

Let $c > 1$ be a constant. To obtain the graph of

- $cf(x)$, stretch G vertically by a factor of c;
- $\frac{1}{c}f(x)$, compress G vertically by a factor of c;
- $f(cx)$, compress G horizontally by a factor of c;
- $f\left(\frac{x}{c}\right)$, stretch G horizontally by a factor of c.

2.4 Transformation of Graphs

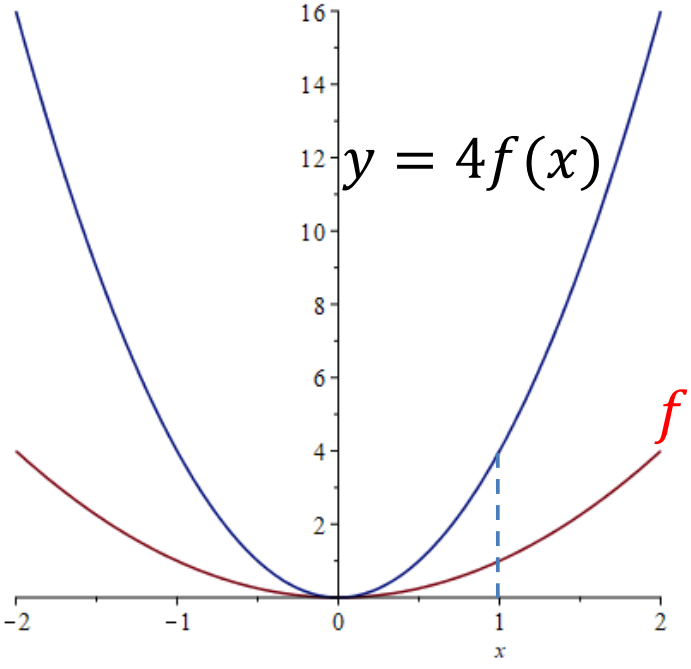


vertical stretch by a factor of 2.

2.4 Transformation of Graphs

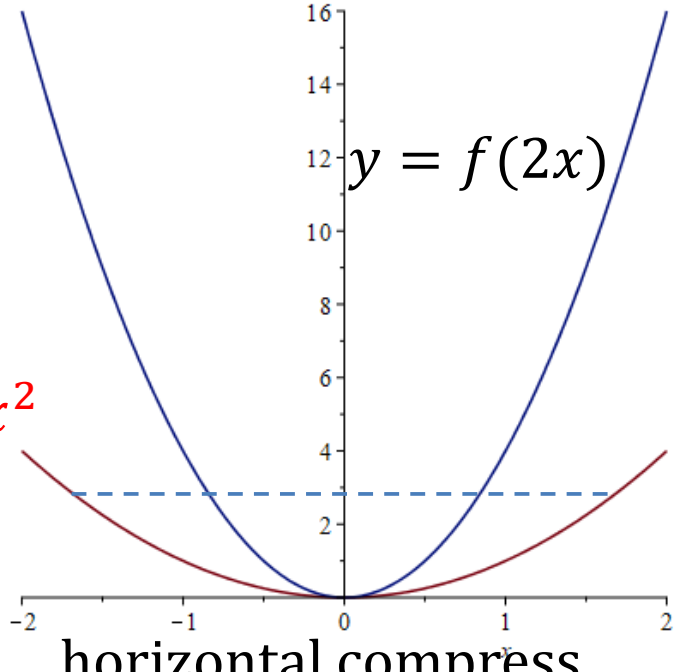
Example Consider $f(x) = x^2$. We have

$$f(2x) = (2x)^2 = 4x^2 = 4f(x).$$



vertical stretch by a factor of 4.

$f(x) = x^2$

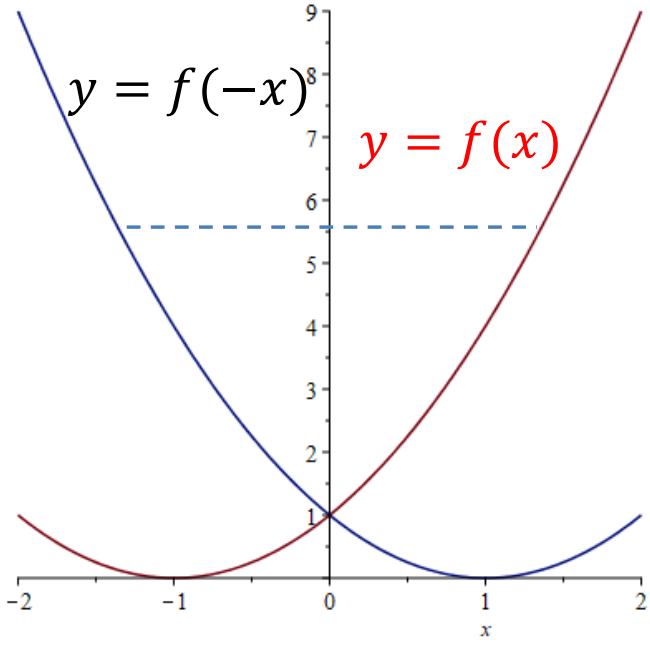


horizontal compress by a factor of 2.

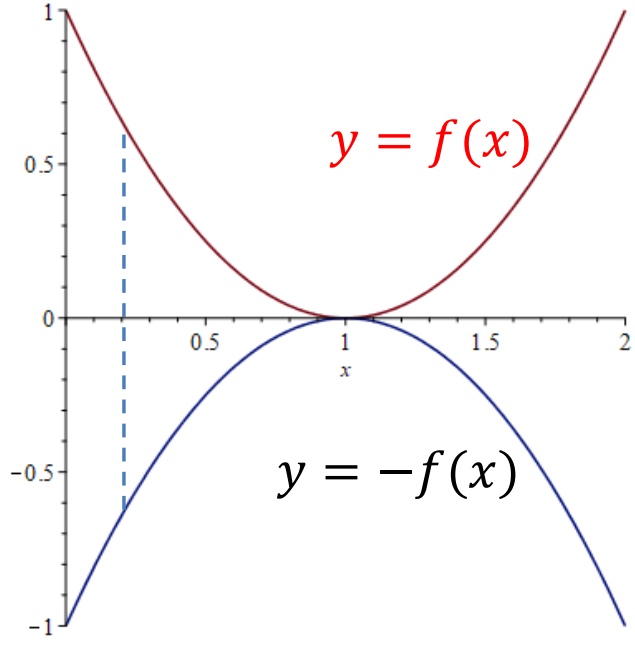
2.4 Vertical and Horizontal Reflections

Reflect the graph of $f(x)$ horizontally about the y-axis to obtain the graph of $f(-x)$.

Reflect the graph of $f(x)$ vertically about the x-axis to obtain the graph of $-f(x)$.



horizontal reflection about the y-axis

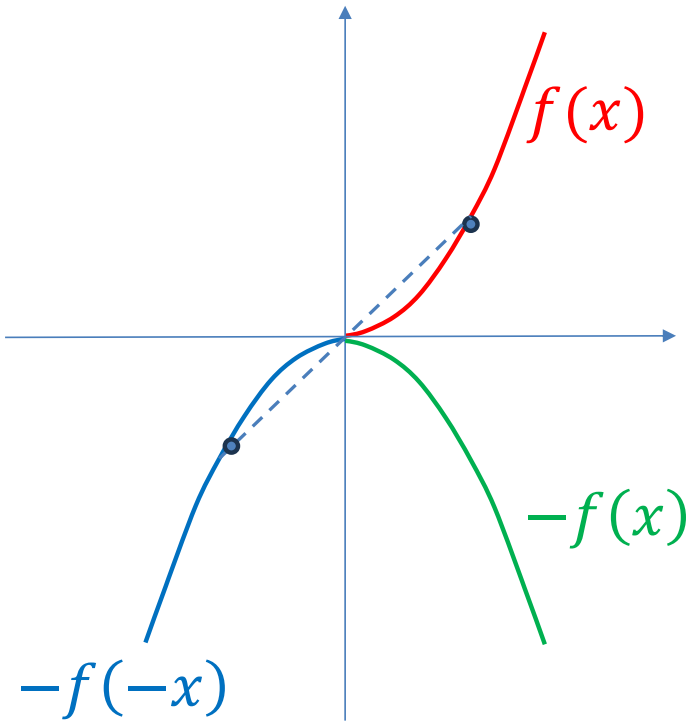


vertical reflection about the x-axis

2.4 Reflection about the Origin

Let G denote the graph of $f(x)$. Reflect G about the origin to obtain the graph of $-f(-x)$. That is, rotate G 180 degree around the origin.

This can also be viewed as: first reflecting G about the x-axis to obtain the graph of $-f(x)$, and then reflecting about the y-axis to obtain the graph of $-f(-x)$.



Reflection about the origin

2.5 Exponential Functions

An *exponential function* has the form a^x, where $a \neq 1$ is a positive constant, which is called the *base* of the exponential function. Its domain is $(-\infty, \infty)$. For example, 2^x, $(1/2)^x$, 10^x are all exponential functions.

Remark Don't confuse exponential functions with power functions x^r.

Recall

$$\left(\frac{1}{2}\right)^x = \frac{1}{2^x} = 2^{-x}.$$

2.5 Laws of Exponents

Let a, b, x and y be real numbers with $a > 0,\ b > 0$. Then

$$
\begin{aligned}
a^0 &= 1, \\
a^{-x} &= \frac{1}{a^x} = \left(\frac{1}{a}\right)^x, \\
a^{x+y} &= a^x a^y, \\
a^{x-y} &= \frac{a^x}{a^y}, \\
(a^x)^y &= a^{xy}, \\
(ab)^x &= a^x b^x, \\
\left(\frac{a}{b}\right)^x &= \frac{a^x}{b^x}.
\end{aligned}
$$

2.5 Laws of Exponents

Examples We have

$$\left(\frac{1}{8}\right)^x = \left(8^{-1}\right)^x = 8^{-x} = \left(2^3\right)^{-x} = 2^{-3x},$$

$$\left(a^{3/4}b^{1/2}\right)^4 = \left(a^{3/4}\right)^4 \left(b^{1/2}\right)^4 = a^3b^2,$$

$$\begin{aligned} 2^{1/4} \times 4^{3/8} &= 2^{1/4} \times \left(2^2\right)^{3/8} \\ &= 2^{1/4} \times 2^{3/4} \\ &= 2^{1/4+3/4} \\ &= 2. \end{aligned}$$

Since $(1/2)^x = 2^{-x}$, the graph of $(1/2)^x$ is the reflection of the graph of 2^x about the y-axis. Thus, we may focus on exponential functions whose base is greater than 1.

2.5 Exponential Functions

